

3.14. Cauchy's Root Test

Statement. If $\sum u_n$ is a positive term series such that

$$\lim_{n \rightarrow \infty} (u_n)^{1/n} = l, \text{ then}$$

(i) the series converges if $l < 1$

(ii) the series diverges if $l > 1$

Proof. By definition of limit, for a given ε , however small, we can find a number m such that

$$| (u_n)^{1/n} - l | < \varepsilon \text{ for } n > m$$

or

$$l - \varepsilon < u_n^{1/n} < l + \varepsilon \text{ for } n > m$$

Omitting the first m terms of the series we have

$$l - \varepsilon < u_n^{1/n} < l + \varepsilon \text{ for all } n \quad \dots(1)$$

Case I. $l < 1$

We can choose ε so small that $l + \varepsilon = k < 1$.

Now $u_n^{1/n} < k$ for all n [From (1)]

$$u_n < k^n \text{ for all } n.$$

$\therefore$

$$u_1 + u_2 + \dots + \infty < k + k^2 + \dots + k^n + \dots \infty$$

$$< \frac{1}{1-k} \text{ (a finite quantity)}$$

$\therefore \sum u_n$ is convergent.

Case II.

$$l > 1$$

We can choose ε so small that $l - \varepsilon > 1$.

$\therefore$

$$u_n^{1/n} > l - \varepsilon \quad [\text{from (1)}]$$

$$\text{i.e., } > 1.$$

$\therefore$

$$u_n > 1$$

Hence $S_n = u_1 + u_2 + u_3 + \dots + u_n > n$

$\therefore$

$$\lim_{n \rightarrow \infty} S_n = \infty$$

$$n \rightarrow \infty$$

or $\sum u_n$ is divergent.

Note. If $\lim_{n \rightarrow \infty} u_n^{1/n} = 1$, the root test fails i.e., gives no information about the nature of the series as regards convergence or divergence.

For example consider the series $\sum \frac{1}{n^p}$

$$\text{Here } \lim_{n \rightarrow \infty} u_n^{1/n} = \lim_{n \rightarrow \infty} \left(\frac{1}{n^p} \right)^{1/n} = \lim_{n \rightarrow \infty} \left(\frac{1}{n^{1/n}} \right)^p = 1 \text{ for all values of } p.$$

But $\sum \frac{1}{n^p}$ converges when $p > 1$ and diverges when $p \leq 1$.

Thus when $\lim_{n \rightarrow \infty} u_n^{1/n} = 1$, we can not say whether $\sum u_n$ is convergent or divergent.

Example. Test the convergence of the series :

$$(i) \quad \sum \frac{1}{n^n}$$

$$(ii) \quad \sum \left(1 + \frac{1}{n} \right)^{n^2} \quad (\text{Rajasthan 1961})$$

$$(iii) \quad \frac{1}{2} + \frac{2}{3}x + \left(\frac{3}{4} \right)^2 x^2 + \left(\frac{4}{5} \right)^3 x^3 + \dots \infty$$

$$(i) \text{ Here } \lim_{n \rightarrow \infty} u_n^{1/n} = \lim_{n \rightarrow \infty} \left(\frac{1}{n^n} \right)^{1/n} \\ = \lim_{n \rightarrow \infty} \frac{1}{n} = 0 \text{ i.e. } < 1.$$

Hence the series is convergent.

$$(ii) \text{ Here } u_n = \left(1 + \frac{1}{n} \right)^{n^2}$$

$$\therefore \lim_{n \rightarrow \infty} u_n^{1/n} = \lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{n} \right)^{n^2} \right]^{1/n} \\ = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e \text{ which is } > 1$$

Hence the series is divergent.

(iii) Omitting the first term we get

$$u_n = \left(\frac{n+1}{n+2} \right)^n x^n$$

$$\therefore \lim_{n \rightarrow \infty} u_n^{1/n} = \lim_{n \rightarrow \infty} \frac{n+1}{n+2} x \\ = x.$$

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Hence the series converges if $x < 1$ and diverges for $x > 1$.

When $x=1$, $u_n = \left(\frac{n+1}{n+2}\right)^n = \frac{(1+1/n)^n}{(1+2/n)^n}$

$$= \frac{\left(1 + \frac{1}{n}\right)^n}{\left[\left(1 + \frac{2}{n}\right)^{n/2}\right]^2}$$

$$\begin{aligned} \therefore \lim_{n \rightarrow \infty} u_n &= \lim_{n \rightarrow \infty} \frac{(1+1/n)^n}{[(1+2/n)^{n/2}]^2} \\ &= \frac{e}{e^2} = \frac{1}{e} (\neq 0) \end{aligned}$$

$\therefore$ the series is divergent.

3.15. Cauchy's Integral Test

Statement. A positive-term series

$$f(1) + f(2) + f(3) + \dots + f(n) + \dots$$

where $f(n)$ does not increase with n , converges or diverges according as the integral

$$\int_1^{\infty} f(x) dx$$

is finite or infinite.

Proof. In the adjoining figure the area under the curve from $x=1$ to $x=n+1$ lies between the sum of the areas of the small rectangles and the sum of the large rectangles, all rectangles having unit bases.

$$\therefore f(1) + f(2) + \dots + f(n)$$

$$\geq \int_1^{n+1} f(x) dx$$

$$\geq f(2) + f(3) + \dots + f(n+1)$$

$$\text{i.e., } S_n \geq \int_1^{n+1} f(x) dx \geq S_{n+1} - f(1)$$

As $n \rightarrow \infty$ we find from the second inequality that if the integral has a finite value then $\lim_{n \rightarrow \infty} S_{n+1}$ is also finite and, therefore $\sum f(n)$ converges.

Similarly if the integral is infinite then we see from the first inequality that $\lim_{n \rightarrow \infty} S_n = \infty$ and, therefore, the series diverges.

Example 1. Apply the integral test to determine the convergence of the p -series.

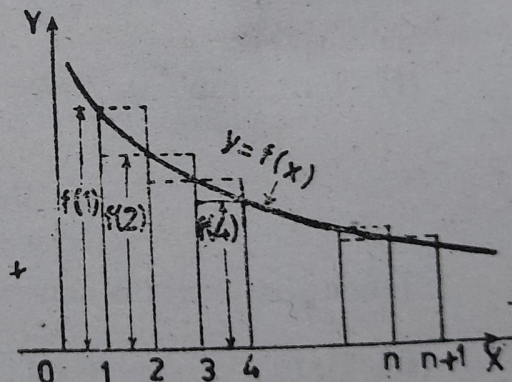


Fig. 3.1

$$\frac{1}{1^p} + \frac{1}{2^p} + \dots + \frac{1}{n^p} + \dots \infty.$$

Here $f(x) = \frac{1}{x^p}$ which decreases as x increases ($x \geq 1$)

$\therefore$ By the integral test, the given series is convergent or divergent according as the integral

$$\int_1^{\infty} \frac{1}{x^p} dx$$

is finite or infinite.

$$\begin{aligned} \text{When } p \neq 1, \int_1^{\infty} \frac{1}{x^p} dx &= \lim_{a \rightarrow \infty} \int_1^a \frac{1}{x^p} dx \\ &= \lim_{a \rightarrow \infty} \left| \frac{x^{1-p}}{1-p} \right|_1^a \\ &= \frac{1}{p-1} \left[\lim_{a \rightarrow \infty} a^{1-p} - 1 \right] \end{aligned}$$

$$\begin{aligned} \text{Now if } p > 1, \frac{1}{1-p} \left[\lim_{a \rightarrow \infty} a^{1-p} - 1 \right] \\ &= \frac{1}{1-p} \left[\lim_{a \rightarrow \infty} \frac{1}{a^{p-1}} - 1 \right] \\ &= \frac{1}{p-1} \text{ which is finite.} \end{aligned}$$

Hence the series converges when $p > 1$

$$\text{When } p < 1, \frac{1}{1-p} \left[\lim_{a \rightarrow \infty} a^{1-p} - 1 \right] \rightarrow \infty.$$

$\therefore$ The series diverges.

$$\text{When } p = 1, \int_1^{\infty} \frac{1}{x} dx = \left| \log x \right|_1^{\infty} = +\infty.$$

$\therefore$ The series diverges.

Thus $\sum \frac{1}{n^p}$ converges when $p > 1$ and diverges when $p \leq 1$.

Example 2. Examine the convergence :

$$(i) \sum_{n=2}^{\infty} \frac{1}{n \log n} \quad (ii) \sum_{n=1}^{\infty} n e^{-n^2}$$

(i) Here $f(x) = \frac{1}{x \log x}$ which decreases as x increases ($x \geq 2$)

$$\text{Now } \int_2^{\infty} \frac{1}{x \log x} dx = \lim_{a \rightarrow \infty} \int_2^a \frac{dx}{x \log x}$$

$$= \lim_{a \rightarrow \infty} \left| \log \log x \right|_2^a = \lim_{a \rightarrow \infty} [\log \log a - \log \log 2] \rightarrow \infty$$

$\therefore$ by integral test the series is divergent.
(ii) Here $f(x) = xe^{-x^2}$ which decreases as x increases.

Now $\int_1^{\infty} xe^{-x^2} dx$

$$= \lim_{a \rightarrow \infty} \int_1^a xe^{-x^2} dx = \lim_{a \rightarrow \infty} \left| -\frac{1}{2} e^{-x^2} \right|_1^a$$

$$= -\frac{1}{2} \lim_{a \rightarrow \infty} (e^{-a^2} - e^{-1}) = \frac{e^{-1}}{2} = \frac{1}{2e}$$

which is finite.

Hence the given series converges.

Exercise 3.4

Examine the convergence of the series using root test :

1. $\sum \left(1 + \frac{1}{\sqrt{n}}\right)^{-n^{3/2}}$

2. $1 + \frac{x}{2} + \frac{x^2}{3^2} + \frac{x^3}{4^3} + \dots \infty \quad (x \geq 0)$

3. (i) $\sum \frac{x^n}{n^n}$ (ii) $\sum \frac{1}{n^n}$ (iv) $\sum \frac{1}{(\log n)^n}$

4. $\frac{2}{1^2}x + \frac{3^2}{2^3}x^2 + \frac{4^3}{3^4}x^3 + \dots + \frac{(n+1)^n}{n^{n+1}}x^n + \dots \infty$

[Hint : $u_n = \left(\frac{n+1}{n}\right)^n \frac{x^n}{n}$

$$\therefore \lim u_n^{1/n} = \lim \frac{n+1}{n} \cdot \frac{x}{n^{1/n}} = x \quad (\because \lim n^{1/n} = 1.)$$

$\therefore$ Series converges when $x < 1$ and diverges when $x > 1$.

For $x=1$, use comparison test with $v_n = \frac{1}{n}$

5. $\left(\frac{2^2}{1^2} - \frac{2}{1}\right)^{-1} + \left(\frac{3^3}{2^3} - \frac{3}{2}\right)^{-2} + \left(\frac{4^4}{3^4} - \frac{4}{3}\right)^{-3} + \dots$

Test for convergence using integral test :

6. $\sum \frac{1}{n (\log n)^2}$

7. $\sum \frac{1}{n^2 + 1}$

8. $\sum \frac{1}{\sqrt{n}}$

Ans. (1) Cogt (2) Cogt (3) All Cogt (5) Cogt (6) Cogt (7) Cogt
(8) Digt

3.16. Of the various tests considered so far, D' Alembert's ratio test is comparatively more useful. In case this test fails (i.e., when $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = 1$), we require other tests to determine

the nature of the series Σu_n . Two such tests will now be developed with the help of another form of comparison test which we establish first.

3.17. Comparison of Ratios. If Σu_n and Σv_n are two positive term series, then

(A) Σu_n converges if (i) Σv_n converges, and (ii) from and after some particular term $\frac{u_{n+1}}{u_n} < \frac{v_{n+1}}{v_n}$.

(B) Σu_n diverges if (i) Σv_n diverges, and (ii) from and after some particular term $\frac{u_{n+1}}{u_n} > \frac{v_{n+1}}{v_n}$.

Let the inequality be satisfied from $n=m+1$ onwards.

Omitting the first m terms of both series, let the series be

$$u_1 + u_2 + u_3 + \dots \text{ and } v_1 + v_2 + v_3 + \dots$$

(A) Since $\frac{u_2}{u_1} < \frac{v_2}{v_1}$, $\frac{u_3}{u_2} < \frac{v_3}{v_2}$, ...

$$\begin{aligned} \therefore u_1 + u_2 + u_3 + \dots &= u_1 \left(1 + \frac{u_2}{u_1} + \frac{u_3}{u_2} \cdot \frac{u_2}{u_1} + \dots \right) \\ &< u_1 \left(1 + \frac{v_2}{v_1} + \frac{v_3}{v_2} \cdot \frac{v_2}{v_1} + \dots \right) \\ &< \frac{u_1}{v_1} (v_1 + v_2 + v_3 + \dots) \end{aligned}$$

Hence if Σv_n is convergent, Σu_n is also convergent.

(B) Proceed as in part (A) above.

Note. The above test may be stated in the following alternative form also.

If Σu_n and Σv_n are two positive term series, then

(A) Σu_n converges if (i) Σv_n converges, and (ii) from and after some particular term $\frac{u_n}{u_{n+1}} > \frac{v_n}{v_{n+1}}$

(B) Σu_n diverges if (i) Σv_n diverges, and (ii) from and after some particular term $\frac{u_n}{u_{n+1}} < \frac{v_n}{v_{n+1}}$

3.18. Raabe's Test (Higher Ratio Test). If $\sum u_n$ is a positive term series such that

$$\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) = l, \text{ then}$$

- (i) the series converges if $l > 1$,
- (ii) diverges if $l < 1$.

Case I : $l > 1$.

Choose p such that $l > p > 1$ and compare the given series $\sum u_n$ with $\sum \frac{1}{n^p}$ which is convergent because $p > 1$.

Now $\sum u_n$ converges if from and after a particular term

$$\frac{u_n}{u_{n+1}} > \frac{(n+1)^p}{n^p} \text{ i.e. } \left(1 + \frac{1}{n} \right)^p$$

$$\text{i.e., if } \frac{u_n}{u_{n+1}} > 1 + \frac{p}{n} + \frac{p(p-1)}{2!} \frac{1}{n^2} + \dots$$

$$\text{i.e., if } n \left(\frac{u_n}{u_{n+1}} - 1 \right) > p + \frac{p(p-1)}{2n} + \dots$$

$$\text{or if } \lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) > p$$

or if $l > p$ which is true because $l > p > 1$.
 $\therefore \sum u_n$ converges if $l > 1$

Case II : $l < 1$

Proceed as in case I above.

Note 1. Raabe's test fails if $l = 1$.

Note 2. (Important) : Raabe's test should be applied only when the D' Alembert's ratio test fails. When Raabe's test also fails further tests are necessary.

It can be proved that in such cases the series is divergent, provided $\frac{u_n}{u_{n+1}}$ does not involve n as an exponent or as a logarithm.

Example 1. Test for convergence :

$$(i) 1 + \frac{1+\alpha}{1+\beta} + \frac{(1+\alpha)(2+\alpha)}{(1+\beta)(2+\beta)} + \dots \infty$$

$$(ii) 1 + \frac{1}{2}x + \frac{1 \cdot 3}{2 \cdot 4}x^2 + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}x^3 + \dots \infty \quad (x > 0)$$

$$(i) \text{ Here } u_n = \frac{(1+\alpha)(2+\alpha)\dots(n-1+\alpha)}{(1+\beta)(2+\beta)\dots(n-1+\beta)}$$

and

$$u_{n+1} = \frac{(1+\alpha)(2+\alpha)\dots(n-1+\alpha)(n+\alpha)}{(1+\beta)(2+\beta)\dots(n-1+\beta)(n+\beta)}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{n+\alpha}{n+\beta} = 1.$$

Thus the Ratio test fails.

$$\begin{aligned}\text{Again } \lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) &= \lim_{n \rightarrow \infty} n \left(\frac{n+\beta}{n+\alpha} - 1 \right) \\ &= \lim_{n \rightarrow \infty} \frac{\beta-\alpha}{1+\alpha/n} \\ &= \beta-\alpha.\end{aligned}$$

Hence by Raabe's test the series converges if $\beta-\alpha > 1$ and diverges if $\beta-\alpha < 1$.

This test also fails if $\beta-\alpha = 1$.

Putting $\beta = 1 + \alpha$, the given series becomes

$$1 + \frac{1+\alpha}{2+\alpha} + \frac{1+\alpha}{3+\alpha} + \frac{1+\alpha}{4+\alpha} + \dots$$

$$\therefore u_n = \frac{1+\alpha}{n+\alpha}.$$

Taking $v_n = \frac{1}{n}$, we have

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{1+\alpha}{n+\alpha} \times n = 1.$$

Since $\sum \frac{1}{n}$ diverges, therefore by comparison test, $\sum u_n$ also diverges.

(ii) Neglecting the first term

$$u_n = \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{2 \cdot 4 \cdot 6 \dots 2n} x^n,$$

$$u_{n+1} = \frac{1 \cdot 3 \cdot 5 \dots (2n-1)(2n+1)}{2 \cdot 4 \cdot 6 \dots 2n \cdot 2n+2} x^{n+1}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{2n+1}{2n+2} \times x = x$$

By ratio test, the series converges if $x < 1$ and diverges if $x > 1$.

For $x=1$, the ratio test fails and we apply Raabe's test and get

$$\begin{aligned}\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) &= \lim_{n \rightarrow \infty} n \left(\frac{2n+2}{2n+1} - 1 \right) \\ &= \lim_{n \rightarrow \infty} n \left(\frac{1}{2n+1} \right) = \frac{1}{2} \text{ which is } < 1.\end{aligned}$$

$\therefore$ The series diverges for $x=1$.

Example 2. Examine the convergence of the series

$$1 + \frac{2^2}{3^2} + \frac{2^2 \cdot 4^2}{3^2 \cdot 5^2} + \frac{2^2 \cdot 4^2 \cdot 6^2}{3^2 \cdot 5^2 \cdot 7^2} + \dots \infty$$

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In this case $u_{n+1} = u_n \cdot \frac{(2n)^2}{(2n+1)^2}$

$$\therefore \lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{(2n)^2}{(2n+1)^2} = 1.$$

Thus the ratio test fails.

Again $\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) = \lim_{n \rightarrow \infty} \left[\frac{(2n+1)^2 - (2n)^2}{(2n)^2} \right]$

$$= \lim_{n \rightarrow \infty} \frac{4n+1}{4n} = 1.$$

$\therefore$ Raabe's test also fails.

Since $\frac{u_n}{u_{n+1}}$ does not involve n as an exponent or as a logarithm, the series diverges (See note 2 Art 3.18).

3.19. Logarithmic Test

Statement. If Σu_n is a positive term series such that

$$\lim_{n \rightarrow \infty} \left(n \log \frac{u_n}{u_{n+1}} \right) = l, \text{ then}$$

(i) the series converges if $l > 1$

(ii) the series diverges if $l < 1$.

Proof. Case I. $l > 1$

Choose p such that $l > p > 1$ and compare Σu_n with $\Sigma \frac{1}{n^p}$ which converges because $p > 1$.

Now Σu_n converges if from and after a particular term

$$\frac{u_n}{u_{n+1}} > \frac{(n+1)^p}{n^p} \text{ i.e., } \left(1 + \frac{1}{n} \right)^p.$$

Taking logarithms of both sides

$$\log \frac{u_n}{u_{n+1}} > p \log \left(1 + \frac{1}{n} \right)$$

$$\text{i.e., if } \log \frac{u_n}{u_{n+1}} > p \left(\frac{1}{n} - \frac{1}{2n^2} + \frac{1}{3n^3} - \dots \right)$$

$$\text{or if } n \log \frac{u_n}{u_{n+1}} > p \left(1 - \frac{1}{2n} + \frac{1}{3n^2} - \dots \right)$$

$$\text{i.e., if } \lim_{n \rightarrow \infty} n \log \frac{u_n}{u_{n+1}} > p$$

i.e., $l > p$ which is true since $l > p > 1$.

Hence Σu_n converges

Case II. $l < 1$

Proceed as in case I above.

Notes. (i) Logarithmic test fails when $l=1$.

(ii) This test is an alternative to Raab's test and should be applied when D' Alembert's ratio test fails and when either (i) n occurs as an exponent in $\frac{u_n}{u_{n+1}}$ or (ii) taking of logarithm of $\frac{u_n}{u_{n+1}}$ makes the evaluation of limits easier.

Example. Test for convergence and divergence the series :

$$(i) 1 + \frac{2x}{2!} + \frac{3^2 x^2}{3!} + \frac{4^3 x^3}{4!} + \frac{5^4 x^4}{5!} + \dots \quad (\text{Roorkee, 1961})$$

$$(ii) x + \frac{2^2 x^2}{2!} + \frac{3^3 x^3}{3!} + \frac{4^4 x^4}{4!} + \frac{5^5 x^5}{5!} + \dots \quad (\text{Roorkee, 1978})$$

$$(i) \text{ Here } u_n = \frac{n^{n-1} x^{n-1}}{n!} \text{ and } u_{n+1} = \frac{(n+1)^n x^n}{(n+1)!}$$

$$\begin{aligned} \text{Therefore } \frac{u_{n+1}}{u_n} &= \frac{(n+1)^n}{n^{n-1}} \cdot \frac{x}{n+1} \\ &= \left(1 + \frac{1}{n}\right)^n \cdot \frac{n}{n+1} x. \end{aligned}$$

and

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = ex.$$

Hence the series is convergent if $ex < 1$, i.e., if $x < \frac{1}{e}$; and is divergent if $x > \frac{1}{e}$. If $x = \frac{1}{e}$, the test fails and we apply logarithmic test.

$$\begin{aligned} \text{Now } \lim_{n \rightarrow \infty} n \log \frac{u_n}{u_{n+1}} &= \lim_{n \rightarrow \infty} n \log \left[\frac{e}{(1 + 1/n)^{n-1}} \right] \\ &= \lim_{n \rightarrow \infty} n \left\{ \log e - (n-1) \log \left(1 + \frac{1}{n} \right) \right\} \\ &= \lim_{n \rightarrow \infty} n \left\{ 1 - \frac{n-1}{n} + \frac{n-1}{2n^2} - \frac{n-1}{3n^3} + \dots \right\} \\ &= \lim_{n \rightarrow \infty} \left\{ n - (n-1) + \left(\frac{1}{2} - \frac{1}{2n} \right) - \frac{(n-1)}{3n^2} + \dots \right\} \\ &= 1 + \frac{1}{2} = \frac{3}{2}, \text{ which is greater than 1.} \end{aligned}$$

Hence the series is convergent for $x = \frac{1}{e}$.

$$(ii) \text{ Here } u_n = \frac{n^n x^n}{n!} \text{ and } u_{n+1} = \frac{(n+1)^{n+1} x^{n+1}}{(n+1)!}$$

$$\text{Therefore } \lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{(n+1)^{n+1} x^{n+1}}{(n+1)!} \times \frac{n!}{n^n x^n}$$

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$$= \lim_{n \rightarrow \infty} \frac{(n+1)^n}{n^n} x = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n x = ex$$

Hence if $ex < 1$, i.e., if $x < \frac{1}{e}$, the series is convergent. If $x > \frac{1}{e}$, the series is divergent.

If $x = \frac{1}{e}$, the ratio test fails and we apply logarithmic test.

We thus have

$$\begin{aligned} \lim_{n \rightarrow \infty} n \log \frac{u_n}{u_{n+1}} &= \lim_{n \rightarrow \infty} n \log \left\{ \left(1 + \frac{1}{n} \right)^{-n} e \right\} \\ &= \lim_{n \rightarrow \infty} n \left\{ \log e - n \log \left(1 + \frac{1}{n} \right) \right\} \\ &= \lim_{n \rightarrow \infty} n \left\{ 1 - n \left(\frac{1}{n} - \frac{1}{2n^2} + \frac{1}{3n^3} + \dots \right) \right\} \\ &= \lim_{n \rightarrow \infty} \left\{ \frac{1}{2} - \frac{1}{3n} + \dots \right\} \\ &= \frac{1}{2} < 1. \end{aligned}$$

Hence the series is divergent if $x = \frac{1}{e}$.

Exercise 3.5

Examine the Following Series for Convergence :

$$1. \quad \sum_{n=1}^{\infty} \frac{1^2 \cdot 5^2 \cdot 9^2 \dots (4n-3)^2}{4^2 \cdot 8^2 \cdot 12^2 \dots (4n)^2}$$

$$2. \quad 1 + a + \frac{a(a+1)}{1 \cdot 2} + \frac{a(a+1)(a+2)}{1 \cdot 2 \cdot 3} + \dots$$

$$3. \quad \frac{2}{7} + \frac{2.5}{7.10} + \frac{2.5.8}{7.10.13} + \dots$$

$$4. \quad \sum \frac{3.6.9 \dots 3n}{7.10.13 \dots (3n+4)} x^n \quad (x > 0)$$

$$5. \quad 1 + \frac{1}{2} \cdot \frac{x^2}{4} + \frac{1.3.5}{2.4.6} \cdot \frac{x^4}{8} + \frac{1.3.5.7.9}{2.4.6.8.10} \cdot \frac{x^6}{12} + \dots$$

$$6. \quad \frac{1^2}{2^2} + \frac{1^2 \cdot 3^2}{2^2 \cdot 4^2} x + \frac{1^2 \cdot 3^2 \cdot 5^2}{2^2 \cdot 4^2 \cdot 6^2} x^2 + \dots$$

$$7. \quad x + \frac{1}{2} \cdot \frac{x^2}{3} + \frac{1.3}{2.4} \cdot \frac{x^5}{5} + \frac{1.3.5}{2.4.6} \cdot \frac{x^7}{7} + \dots$$

$$8. \quad \frac{a+x}{1} + \frac{(a+2x)^2}{2!} + \frac{(a+3x)^3}{3!} + \dots$$

$$9. \quad 1 + \frac{1!}{2}x + \frac{2!}{3^2}x^2 + \frac{3!}{4^3}x^3 + \dots$$

10. (Hypergeometric series)

$$1 + \frac{\alpha \cdot \beta}{1 \cdot \gamma} x + \frac{\alpha(\alpha+1)\beta(\beta+1)}{1 \cdot 2 \cdot \gamma(\gamma+1)} x^2 + \\ + \frac{\alpha(\alpha+1)(\alpha+2)\beta(\beta+1)(\beta+2)}{1 \cdot 2 \cdot 3 \cdot \gamma(\gamma+1)(\gamma+2)} x^3 + \dots$$

(Hint. Neglecting first term

$$u_{n+1} = u_n \frac{(\alpha+n)(\beta+n)}{(n+1)(\gamma+n)} x$$

$\therefore \lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = x$. Hence the series converges if $x < 1$ and

diverges if $x > 1$.

When $x=1$, ratio test fails.

Applying Raabe's test

$$\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) = \lim_{n \rightarrow \infty} n \left[\frac{(n+1)(n+\gamma)}{(n+\alpha)(n+\beta)} - 1 \right] \\ = \lim_{n \rightarrow \infty} n \left[\frac{n(1+\gamma-\alpha-\beta) + \gamma - \alpha\beta}{n^2 + n(\alpha+\beta) + \alpha\beta} \right] \\ = 1 + \gamma - \alpha - \beta$$

$\therefore$ The series converges when $1 + \gamma - \alpha - \beta > 1$

i.e., $\gamma - \alpha - \beta > 0$ and diverges when $\gamma - \alpha - \beta < 0$.

When $\gamma - \alpha - \beta = 0$, Raabe's test fails.

Since $\frac{u_n}{u_{n+1}}$ does not involve n as an exponent or as a logarithm, therefore the series is divergent when $\gamma - \alpha - \beta = 0$.

(See Art 3.18 note 2)]

Ans. (1) Cogt (2) Cogt for $a \leq 0$, Digt for $a > 0$ (3) Cogt (4) Cogt for $x \leq 1$, Digt for $x > 1$ (5) Cogt for $x^2 \leq 1$, Digt for $x^2 > 1$ (6) Cogt for $x < 1$, Digt for $x \geq 1$ (7) Cogt for $x \leq 1$, Digt for $x > 1$ (8) Cogt for $x < \frac{1}{e}$, Digt for $x \geq \frac{1}{e}$ (9) Cogt for $x < e$, Digt for $x \geq e$.

3.20. Alternating Series.

Def. A series whose terms are alternately positive and negative is called an alternating series.

The general form of such a series is

$$u_1 - u_2 + u_3 - u_4 + \dots (u_n > 0)$$

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Thus $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$ is an alternating series.

Leibnitz Test for Alternating Series

(Delhi, 1983)

If the alternating series

$$u_1 - u_2 + u_3 - u_4 + \dots \quad (u_n > 0)$$

is such that

(i) $u_1 > u_2 > u_3 > \dots$

(ii) $\lim_{n \rightarrow \infty} u_n = 0$

then the series converges.

Proof. Consider S_{2n} the sum of an even number of terms.

$$S_{2n} = (u_1 - u_2) + (u_3 - u_4) + \dots + (u_{2n-1} - u_{2n}) \quad \dots(1)$$

$$\text{Also } S_{2n} = u_1 - (u_2 - u_3) - (u_4 - u_5) - \dots - u_{2n}$$

$$= u_1 - [(u_2 - u_3) + (u_4 - u_5) + \dots + u_{2n}] \quad \dots(2)$$

From (1) S_{2n} is positive and increases with n .

($\because u_1 > u_2 > u_3 > \dots$ and hence $u_1 - u_2, u_3 - u_4$ etc are all +ve)

From (2) S_{2n} is always less than u_1 ,

$\therefore S_{2n}$ tends to a finite limit as $n \rightarrow \infty$.

Now consider S_{2n+1} , the sum of odd number of terms.

$$\text{Since } S_{2n+1} = S_{2n} + u_{2n+1},$$

$$\text{therefore } \lim_{n \rightarrow \infty} S_{2n+1} = \lim_{n \rightarrow \infty} S_{2n} + \lim_{n \rightarrow \infty} u_{2n+1} \quad \dots(3)$$

$$= \lim_{n \rightarrow \infty} S_{2n} \quad (\because \lim_{n \rightarrow \infty} u_n = 0)$$

Thus S_n tends to the same limit whether n is even or odd.

Hence the given series is convergent.

Note. It should be noted that $\lim_{n \rightarrow \infty} u_n = 0$ is necessary for convergence. If $\lim_{n \rightarrow \infty} u_n \neq 0$ then from (3), S_{2n} and S_{2n+1} tend to different limits and therefore the series is oscillatory.

Example 1. Show that the series

$$1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots \text{converges.}$$

If the given series is $u_1 - u_2 + u_3 - u_4 + \dots$

then $|u_n| = \frac{1}{n^2}$ and $|u_{n+1}| = \frac{1}{(n+1)^2}$.

Clearly $\frac{1}{n^2} > \frac{1}{(n+1)^2}$

Also $\lim_{n \rightarrow \infty} |u_n| = \lim_{n \rightarrow \infty} \frac{1}{n^2} = 0$

$\therefore$ By Leibnitz test, the series converges.

Example 2. Discuss the convergence of the series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1} n}{2n-1}$$

Let the given series be denoted by

$$u_1 - u_2 + u_3 - u_4 + \dots$$

$$\begin{aligned} \text{Now } |u_n| - |u_{n+1}| &= \frac{n}{2n-1} - \frac{n+1}{2n+1} \\ &= \frac{1}{(2n-1)(2n+1)} > 0 \end{aligned}$$

$\therefore |u_n| > |u_{n+1}|$ for all n .

$$\text{But } \lim_{n \rightarrow \infty} |u_n| = \lim_{n \rightarrow \infty} \frac{n}{2n-1} = \frac{1}{2} \neq 0.$$

$\therefore$ The alternating series is not convergent. It is oscillatory.

3.21. Absolute Convergence, Conditional Convergence

Let $\Sigma u_n = u_1 + u_2 + u_3 + \dots$... (1)

be a series in which any term may be positive or negative.

If $|u_n|$ denotes the absolute value of u_n i.e.,

$|u_n| = u_n$ if u_n is positive and $|u_n| = -u_n$ if u_n is negative,
then $\Sigma |u_n| = |u_1| + |u_2| + |u_3| + \dots$

is a positive term series each term of which is numerically equal to the corresponding term of (1).

For example if

$$\Sigma u_n = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \quad \dots (2)$$

$$\text{then } \Sigma |u_n| = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots \quad \dots (3)$$

Here Σu_n is convergent but $\Sigma |u_n|$ is divergent.

$$\text{Again if } \Sigma u_n = 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots \quad \dots (4)$$

$$\text{then } \Sigma |u_n| = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots \quad \dots (5)$$

In this case both Σu_n and $\Sigma |u_n|$ are convergent.

Thus if Σu_n is convergent, $\Sigma |u_n|$ may be convergent or divergent.

To distinguish between the two cases that arise in this way, we have the following definitions

Def. (a) A series Σu_n which contains positive as well as negative terms is said to be absolutely convergent if $\Sigma |u_n|$ is convergent.

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(b) A series $\sum u_n$ is said to be conditionally convergent if $\sum u_n$ is convergent but $\sum |u_n|$ is divergent.

Thus the series (2) is conditionally convergent while (4) is absolutely convergent.

It should be noted that by property (2) Art 3.4, if $\sum |u_n|$ is convergent then $\sum u_n$ is also convergent.

Example. Test the following series for convergence and absolute convergence :

(i) $1 - \frac{1}{2\sqrt{2}} + \frac{1}{3\sqrt{3}} - \frac{1}{4\sqrt{4}} + \dots$ (Roorkee, 1971)

(ii) $1 - 2x + 3x^2 - 4x^3 + \dots$ (Roorkee 1972)

(i) Here the terms are alternately positive and negative, each term is numerically less than the preceding term, and

$$\lim |u_n| = \lim \left(\frac{1}{n\sqrt{n}} \right) = 0.$$

Hence the series is convergent.

Also $\sum |u_n| = \sum \frac{1}{n^{3/2}}$, which is convergent. Hence the given series is absolutely convergent.

(ii) Applying ratio test to $\sum |u_n|$, we get

$$\lim \frac{|u_{n+1}|}{|u_n|} = \lim \frac{n+1}{n} |x| = |x|.$$

Hence the given series is absolutely convergent (and therefore convergent) if $|x| < 1$.

If $|x| \geq 1$, u_n increases numerically as n increases. Hence the series will be oscillatory if $x \geq 1$. If $x \leq -1$, the terms will be all positive and the series will be divergent.

Exercise 3.6

Test the convergence of the series :

1. $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$

(Delhi 1982)

2. $1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + \dots$

3. $\frac{1}{1^p} - \frac{1}{2^p} + \frac{1}{3^p} - \frac{1}{4^p} + \dots$

4. $\sum \frac{(-1)^n n}{2n+3}$

5. $\frac{1}{n} - \frac{1}{n+a} + \frac{1}{n+2a} - \frac{1}{n+3a} + \dots$

($a > 0$)

Examine the following convergent series for absolute or conditional convergence.

6. $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \dots$

7. $1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + \dots$

8. $1 - \frac{2}{3} + \frac{3}{3^2} - \frac{4}{3^3} + \dots$

Ans. (1) cogt (2) cogt (3) cogt for $p > 0$ (4) digt (5) cogt (6) Abs. cogt (7) Cond. cogt (8) Abs. cogt.

3.22. Summary of the Procedure for Testing a Series for Convergence

The series to be tested may be classified as follows :

- I. Alternating series
- II. Positive term series of constants
- III. Positive term power series

Type I. Apply Leibnitz test

Type II. Find $\lim u_n$.

If $\lim u_n \neq 0$, the series diverges.

If $\lim u_n = 0$, then

(i) u_n may be an algebraic fraction in n . In this case u_n may be comparable with $\frac{1}{n^p}$. Apply comparison test.

(ii) If n occurs as exponent of u_n , apply root test.

(iii) In case above tests are not applicable, apply D' Alembert's ratio test.

(iv) When ratio test fails, apply Raabes' test. If this test also fails but u_n is an algebraic function of n then the series diverges.

Apply Logarithmic test as an alternative to Raabes' test when n occurs as an exponent of $\frac{u_n}{u_{n+1}}$

(v) Integral test is an alternative to comparison and ratio tests when the integral involved can be easily evaluated.

Type III. Apply ratio test. If $\lim \frac{u_{n+1}}{u_n} = kx$ (k a constant) then the series converges if $x < \frac{1}{k}$ and diverges if $x > \frac{1}{k}$.

For $x = \frac{1}{k}$, proceed as in Type II above.